

Breast Cancer DRFS Prediction Model v1.0

Overview

This project uses machine learning to predict distant relapse-free survival (DRFS) outcomes in breast cancer patients using publicly available gene expression data from the Gene Expression Omnibus (GEO) database.

The workflow was developed in Python using Google Colab and applies a logistic regression classifier to microarray gene expression data from GEO dataset GSE25055. The project focuses on building a reproducible bioinformatics pipeline that processes raw biological data, performs feature selection and normalization, trains a predictive model, and evaluates model performance using multiple metrics and visualizations.

Dataset Information

Primary Dataset

- Dataset: GSE25055
- Source: NCBI GEO Database
- URL: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE25055>

*Important Note

The original training dataset was not uploaded directly to this repository due to file size limitations and GEO distribution policies. Instead, the dataset can be downloaded directly from the GEO database using the link above. This dataset was split within the code to form training and testing data.

The workflow expects the GEO SOFT file:

```
``` id="5vkgfa"  
GSE25055_family.soft.gz
```

This SOFT can be found from the URL provided

### **Project Goals**

The main goals of this project were to:

1. Learn how publicly available biomedical datasets are used in computational biology research
2. Build a full machine learning workflow from raw biological data
3. Explore how gene expression patterns can predict cancer outcomes
4. Develop reproducible research documentation using BioCompute Objects (BCOs)

## **Workflow Summary**

### **1. Data Loading**

The GEO dataset is loaded using the GEOparse Python library.

### **2. Metadata Extraction**

Clinical metadata is extracted from the SOFT file, including the DRFS outcome variable.

### **3. Expression Matrix Processing**

Gene expression values are extracted and reorganized into a samples x features matrix.

### **4. Probe-to-Gene Mapping**

Microarray probe IDs are mapped to biologically meaningful gene symbols using GPL96 annotation data.

### **5. Data Cleaning and Matching**

Expression data and metadata are aligned using matching sample identifiers.

### **6. Feature Selection**

The top 1000 genes with the highest variance are selected to reduce dimensionality and improve model performance.

### **7. Data Normalization**

Features are standardized using StandardScaler from scikit-learn.

### **8. Model Training**

A logistic regression classifier is trained using stratified train/test splitting.

### **9. Model Evaluation**

Performance metrics include:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

## 10. Visualization

The workflow generates:

- ROC curve
- Precision-recall curve
- Confusion matrix

## 11. Biological Interpretation

The most important model features are analyzed and mapped back to genes to investigate their potential biological relevance in cancer progression and treatment response.

## Future Directions

This project will continue as an independent research effort over the summer. Planned next steps include:

### 1. Expanding the Dataset

Additional GEO breast cancer datasets may be integrated to improve model generalizability and increase sample size.

### 2. Testing Advanced Models

Future versions may compare logistic regression against:

- Random forests
- XGBoost
- Neural networks
- Ensemble methods

### 3. Improved Biological Analysis

Future work may include pathway analysis and deeper investigation into biologically significant genes identified by the model.

### 4. Reproducibility and Documentation

The workflow is being documented using a BioCompute Object (BCO) to support reproducibility and transparent computational research practices.

## Technologies Used

- Python
- Google Colab
- GEParse
- pandas
- numpy
- scikit-learn
- matplotlib